

ROMAN, Romania

WMO: 151110

Lat: 46.9692N

Lon: 26.9119E

Elev: 217

StdP: 98.74

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.9	-16.9	0.9	-14.8	-14.2	1.1	-12.2	13.7	-5.8	11.7	-6.9	1.3	0	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.3	20.6	29.8	20.1	28.3	19.5	22.4	28.2	21.5	27.2	20.7	26.2	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	24.5	19.7	14.8	23.7	18.8	14.0	22.8	66.9	28.5	63.5	27.2	60.6	26.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.3	7.3	-19.6	34.0	3.7	1.6	-22.3	35.2	-24.5	36.2	-26.5	37.1	-29.2	38.3		
(5)			-19.8	24.1	3.7	1.2	-22.5	25.0	-24.6	25.7	-26.7	26.4	-29.4	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-3.0	0.0	4.2	10.4	15.8	19.9	21.0	20.8	16.2	9.9	3.9	-1.4	(6)
(7)		DBStd	9.56	4.63	5.42	4.90	4.25	3.60	2.91	2.60	2.99	4.25	4.18	5.18	5.47	(7)
(8)		HDD10.0	1518	402	281	183	47	4	0	0	0	2	53	191	354	(8)
(9)		HDD18.3	3373	660	514	437	237	92	18	8	12	90	261	432	612	(9)
(10)		CDD10.0	1469	0	1	5	60	184	296	341	334	188	51	9	0	(10)
(11)		CDD18.3	282	0	0	0	1	14	63	90	88	25	1	0	0	(11)
(12)		CDH23.3	2593	0	0	0	13	162	544	758	849	258	8	0	0	(12)
(13)		CDH26.7	754	0	0	0	1	24	133	222	306	68	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.9	2.9	3.3	3.4	2.8	2.6	2.2	2.4	2.4	2.7	2.8	(14)	
(15)	Precipitation	PrecAvg	505	19	18	21	48	68	81	79	57	39	26	27	22	(15)
(16)		PrecMax	700	90	50	68	102	157	222	182	154	189	110	59	65	(16)
(17)		PrecMin	319	2	1	0	10	12	19	24	10	0	3	2	1	(17)
(18)		PrecStd	110	16	14	16	26	38	40	34	37	40	25	16	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	15.3	20.0	24.7	29.0	31.5	32.9	33.5	30.9	24.6	19.0	11.6	(19)
(20)			MCWB	5.1	9.6	11.3	15.3	17.9	22.0	21.5	21.0	19.0	16.6	13.1	8.5	(20)
(21)		2%	DB	5.7	11.3	16.1	22.6	27.0	29.6	30.8	31.6	28.4	21.5	15.5	9.1	(21)
(22)			MCWB	4.1	7.9	9.7	14.0	17.7	20.7	21.1	20.1	17.7	15.2	11.5	6.8	(22)
(23)		5%	DB	4.2	8.4	13.6	20.4	25.1	28.2	29.1	30.1	26.4	18.9	12.9	6.5	(23)
(24)			MCWB	3.0	6.0	8.2	12.8	16.9	19.9	20.4	19.7	17.0	14.1	10.5	4.9	(24)
(25)		10%	DB	2.6	6.3	11.4	18.1	23.2	26.8	27.5	28.1	24.0	16.6	10.7	4.5	(25)
(26)			MCWB	1.7	4.7	7.1	11.9	15.8	19.4	20.0	19.4	16.4	13.0	9.0	3.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	10.1	11.9	16.1	20.5	23.6	23.8	23.6	20.1	18.1	13.9	9.5	(27)
(28)			MCDWB	6.7	14.9	18.5	23.8	26.5	29.3	28.7	30.4	29.1	23.2	17.6	11.3	(28)
(29)		2%	WB	4.5	7.8	10.1	14.7	18.7	22.1	22.3	22.1	18.9	16.2	12.2	7.0	(29)
(30)			MCDWB	5.4	10.6	15.3	21.1	24.3	27.8	27.8	28.4	25.2	19.8	15.2	9.0	(30)
(31)		5%	WB	3.0	6.4	8.8	13.4	17.7	21.1	21.5	20.9	17.9	14.8	10.8	5.2	(31)
(32)			MCDWB	4.0	8.3	12.9	18.9	23.6	26.5	26.9	27.2	24.0	18.0	12.6	6.5	(32)
(33)		10%	WB	1.7	5.0	7.5	12.2	16.6	20.2	20.7	20.0	17.0	13.2	9.3	3.5	(33)
(34)			MCDWB	2.5	6.2	10.5	17.2	21.8	25.2	25.9	26.4	22.8	16.1	10.7	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	7.0	8.9	10.3	11.0	10.9	11.2	12.0	11.0	9.5	6.3	5.8	(35)
(36)			MCDBR	7.6	10.4	13.5	14.7	14.3	13.4	14.1	15.5	14.8	13.7	10.8	8.4	(36)
(37)			MCWBR	5.7	6.8	7.7	7.2	6.3	5.6	5.6	5.6	5.5	7.7	6.6	6.4	(37)
(38)		5% WB	MCDBR	6.8	9.9	12.5	13.2	12.9	12.3	12.1	13.1	13.1	12.3	9.6	8.1	(38)
(39)			MCWBR	5.4	6.9	7.5	6.8	6.2	5.9	5.7	5.5	5.4	7.2	6.2	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.350	0.388	0.448	0.421	0.418	0.444	0.452	0.408	0.373	0.345	0.318	(40)	
(41)		taud	2.297	2.245	2.211	2.076	2.230	2.273	2.208	2.177	2.260	2.340	2.358	2.334	(41)	
(42)		Ebn at Noon	750	801	824	805	846	849	819	793	795	768	716	706	(42)	
(43)		Edh at Noon	77	100	121	152	135	130	137	136	114	91	74	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.23	2.00	3.19	4.40	5.63	5.91	5.95	5.39	3.81	2.44	1.27	0.97	(44)	
(45)		RadStd	0.17	0.27	0.33	0.39	0.47	0.52	0.47	0.41	0.42	0.29	0.11	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.67	N/A	N/A	+0.79	N/A	N/A	-140	-200	+101	+38

Nomenclature: See separate page